



S5 2026 TEST 1 : Dimensional Analysis

Time 1 hour

QUESTION 1

During a university engineering competition, a student proposed a new equation for calculating the power output of a small hydroelectric turbine they had designed. The equation combined variables like water density, flow rate, gravitational acceleration, and turbine radius. Before building a prototype, the team's physics advisor suggested they check the dimensional consistency of the proposed equation to avoid fundamental flaws in their design logic. This step would save them considerable time and resources by ensuring the mathematical relationship between the physical quantities made sense before moving to the experimental phase.

Task:

As a learner of physics;

(a) State the fundamental reason why checking dimensional consistency is a crucial step in formulating physical equations.

(b) The proposed equation for power is $P = k\rho Av^3$, where ρ is density, A is area, and v is velocity.

Determine the dimensions of the constant k .

(c) Explain what the result of your dimensional analysis implies about the physical validity of the student's equation.

QUESTION 2

In a physics laboratory, a student derives an expression for the period T of a simple pendulum. From their observations, they suggest that the period might be related to the mass of the bob (m), the length of the string (l), and the acceleration due to gravity (g) by the equation $T = 2\pi m^a l^b g^c$. Their teacher informs them that one of these variables does not actually affect the period and that dimensional analysis can help identify which one and verify the correct relationship.

Task:

As a learner of physics;

(a) Using dimensional analysis, show that the period T cannot depend on the mass m of the bob.

(b) Using the dimensions of l and g , determine the values of exponents a , b and c that make the equation dimensionally consistent.

END